

MTH344 Sample Quiz 2 (100 pts + EC 20 pts)

Please print your name clearly. A penalty may be applied if your name is difficult to read.

Last Name: _____

First Name: _____

Instructions:

- There are six problems. The points for each problem are indicated next to the question.
- To earn full credit, you must show all necessary work. If it is unclear which assumptions or theorems may be used without proof, write out as much of your reasoning as possible.
- This test is closed-notes, and no computing devices (including calculators) are allowed.

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P1 (20 pts)

Let $X_i \stackrel{\text{i.i.d.}}{\sim} \text{Gamma}(\alpha, \lambda)$ where $\theta = (\alpha, \lambda)^T$. Show that $\mathbf{T}(\mathbf{X}) = (\sum_{i=1}^n X_i, \prod_{i=1}^n X_i)^T$ is a sufficient statistic for θ .

P2 (20 pts)

The PDF of a *Weibull* with $\theta = (k, \lambda)^T$ where $k > 0$ and $\lambda > 0$ is given by:

$$f(x; \theta) = \frac{k}{\lambda} \left(\frac{x}{\lambda}\right)^{k-1} \exp\left(-\left(\frac{x}{\lambda}\right)^k\right), \quad x \geq 0. \quad (1)$$

Let $X_1, \dots, X_n$ be i.i.d. observations from a Weibull distribution with a known k and an unknown λ . Suppose we test the following simple hypotheses:

$$H_0 : \theta = \theta_0 = (k, \lambda_0)^T \quad \text{vs.} \quad H_1 : \theta = \theta_1 = (k, \lambda_1)^T, \quad (2)$$

where $\lambda_0 < \lambda_1$.

a. (10 pts) Assuming n is sufficiently large, apply the CLT to derive the likelihood ratio test at a significance level of α , where $0 < \alpha < 1$.

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b. (10 pts) Based on the test derived in part a), determine the minimum sample size n required to guarantee a target power of at least $1 - \beta$.

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P3 (20 pts)

Let $X_1, \dots, X_n$ be i.i.d. random variables with PDF $f(x; \theta_0)$ where $\theta_0 \in \Omega$. Denoting $\bar{l}_n(\theta) = n^{-1} \sum_{i=1}^n \log f(X_i; \theta)$ and $\bar{l}_\infty(\theta) = E_{\theta_0}[\log f(X; \theta)]$, we assume the following regularity conditions:

- (R1) (Identifiability) For any $\theta \in \Omega$, if $f(\cdot; \theta) = f(\cdot; \theta_0)$, then $\theta = \theta_0$.
- (R2) (Continuity) $\log f(x; \theta)$ is continuous with respect to θ for all x .
- (R3) (Compactness) The parameter space Ω is compact.
- (R4) (Uniform Convergence) $\sup_{\theta \in \Omega} |\bar{l}_n(\theta) - \bar{l}_\infty(\theta)| \xrightarrow{P} 0$. That is, for any $\epsilon > 0$,

$$\lim_{n \rightarrow \infty} P_{\theta_0} \left(\sup_{\theta \in \Omega} |\bar{l}_n(\theta) - \bar{l}_\infty(\theta)| \geq \epsilon \right) = 0. \quad (3)$$

- a. (10 pts) Show that, for any $\epsilon > 0$, $P_{\theta_0} \left(|\hat{\theta}_n^{\text{MLE}} - \theta_0| \geq \epsilon \right) \leq P_{\theta_0} \left(\sup_{\theta \in \Omega} |\bar{l}_n(\theta) - \bar{l}_\infty(\theta)| \geq \delta(\epsilon)/2 \right)$.

Here, $\delta(\epsilon) = \bar{l}_\infty(\theta_0) - \sup_{\theta \in \Omega: |\theta - \theta_0| \geq \epsilon} \bar{l}_\infty(\theta)$.

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b. (10 pts) Consider a random sample $X_1, \dots, X_n \stackrel{i.i.d.}{\sim} \mathcal{U}(0, \theta)$ where the true parameter is $\theta_0 > 0$ and the parameter space is $\Omega = (0, \infty)$, which is non-compact. Show that, the regularity condition (R2) is also violated, and that even in this case, $\hat{\theta}_n^{\text{MLE}} \xrightarrow{p} \theta_0$.

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P4 (20 pts)

Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with unknown μ and known σ^2 . We test $H_0 : \mu = \mu_0$ vs. $H_1 : \mu \neq \mu_0$.

a. (10 pts) Let k_1, k_2 be constants, and define:

$$d_{k_1, k_2}^*(\mathbf{X}) = I \left(\frac{f_{\mu_1}(\mathbf{X})}{f_{\mu_0}(\mathbf{X})} > k_1 + k_2 \frac{n(\bar{X} - \mu_0)}{\sigma^2} \right). \quad (4)$$

Show that for any unbiased test $d(\mathbf{X})$ of level α and any given $\mu_1 \neq \mu_0$, the following holds:

$$\mathbb{E}_{\mu_1}[d(\mathbf{X})] \leq \mathbb{E}_{\mu_1}[d_{k_1, k_2}^*(\mathbf{X})] + k_1 (\alpha - \mathbb{E}_{\mu_0}[d_{k_1, k_2}^*(\mathbf{X})]) - k_2 \mathbb{E}_{\mu_0} \left[d_{k_1, k_2}^*(\mathbf{X}) \frac{n(\bar{X} - \mu_0)}{\sigma^2} \right]. \quad (5)$$

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b. (10 pts) Show that, with an appropriately chosen $(k_1, k_2)^T$,

$$d_{k_1, k_2}^*(\mathbf{X}) = I \left(\frac{\sqrt{n}|\bar{X} - \mu_0|}{\sigma} > z_{\alpha/2} \right). \quad (6)$$

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P5 (20 pts)

Let $(X, Y)^T$ follow a *logistic model* (or *logit model*):

$$f_{(X,Y)}(x, y; \theta) = f_X(x) \text{Ber}(y; p(x; \theta)) \quad (7)$$

where $\text{Ber}(y; p)$ denotes the PMF of a Bernoulli distribution with parameter p , $\theta = (\beta_0, \beta_1)^T$, and $f_X(x)$ is the marginal distribution of X , which does not depend on θ . The success probability is defined by the logistic function:

$$p(x; \theta) = \frac{1}{1 + \exp(-(\beta_0 + \beta_1 x))}. \quad (8)$$

a. (10 pts) Let $\tilde{\mathbf{X}}$ be the $n \times 2$ observed *design matrix*:

$$\tilde{\mathbf{X}} = \begin{pmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix}. \quad (9)$$

Show that $\mathcal{I}_n(\theta \mid \tilde{\mathbf{X}}) := \text{Var}_\theta[\nabla_\theta l_n(\theta) \mid \tilde{\mathbf{X}}]$ is given by:

$$\mathcal{I}_n(\theta) = \tilde{\mathbf{X}}^T \mathbf{W} \tilde{\mathbf{X}} \quad (10)$$

where $\mathbf{W}$ is an $n \times n$ diagonal matrix with elements $W_{ii} = p(x_i; \theta)(1 - p(x_i; \theta))$.

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b. (10 pts) Show that as the sample variance of $x_1, \dots, x_n$ approaches zero, the determinant of the conditional Fisher information from a) converges to zero.

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P6 (EC 20 pts)

Suppose $(X_i, Y_i)^\top$ for $i = 1, \dots, n$ are i.i.d. random vectors from a bivariate normal distribution with parameter $\theta = (\mu_X, \mu_Y, \sigma_X^2, \sigma_Y^2, \rho)^\top$. The parameter space is given by:

$$\Omega = \{\theta : \mu_X, \mu_Y \in \mathbb{R}, \sigma_X^2, \sigma_Y^2 > 0, -1 < \rho < 1\}.$$

a. (10 pts) Show that the MLE over Ω is given by:

$$\hat{\theta} = \left(\bar{X}, \bar{Y}, \frac{(n-1)S_X^2}{n}, \frac{(n-1)S_Y^2}{n}, R \right)^T, \quad (11)$$

where R denotes the sample correlation coefficient.

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b. (10 pts) Show that $\sqrt{n}(\hat{\rho} - \rho) \xrightarrow{d} \mathcal{N}(0, (1 - \rho^2)^2)$.